

RAMESH DAS

Senior Full Stack Engineer (SDE III) — Node.js | React.js | Go | AWS | GenAI & Agentic AI | Microservices | System Design

Delhi, India | +91 7059476213 | rameshdas028@gmail.com | LinkedIn

PROFESSIONAL SUMMARY

Full Stack Engineer with **7+ years** building high-availability distributed systems on Node.js, React.js, and AWS, with hands-on experience shipping production GenAI/Agentic AI solutions (RAG pipelines, MCP servers, LLM-based automation). Track record of cutting infrastructure cost (**~50%**) and P99 latency (**~60%**) while scaling systems to **100K+ RPS** and multi-million-record data pipelines. Comfortable owning a system end-to-end — architecture, cloud infra, delivery, and post-launch optimization.

CORE SKILLS

Languages: JavaScript (ES6+), TypeScript, Go

Backend: Node.js, Express.js, REST & GraphQL API Design, gRPC

Architecture: Microservices, Event-Driven, Distributed Systems, System Design

Cloud & DevOps: AWS (Solutions Architect), Terraform, Docker, Kubernetes, GitHub Actions, Jenkins, Serverless

Frontend: React.js, Vue.js, Redux Toolkit, Microfrontends

AI / GenAI: LLMs, AI Agents, Agentic AI, RAG, Prompt Engineering, MCP

Databases: MySQL, MongoDB, DynamoDB, Redis, Elasticsearch

Monitoring: CloudWatch, Prometheus, Grafana, Loki

PROFESSIONAL EXPERIENCE

Full Stack Developer | SUTRA AI Private Limited (parent of String Ventures) — Noida, India *Nov 2022 – Present*

- Built production GenAI/Agentic AI systems — RAG pipelines, MCP servers, and LLM-based agent workflows — that automate product queries, order tracking, and delivery support for **24/7 customer service**.
- Built an AI-powered document extraction pipeline (LLMs + OCR) converting unstructured documents into structured JSON, cutting manual processing effort by **90%**.
- Designed high-throughput data ingestion pipelines processing **80K–1M records**, reducing processing time from ~1 hour to **under 5 minutes**.
- Optimized backend APIs, queries, and caching, cutting P90 latency from ~800ms to **~300ms** and P99 latency from ~1.5s to **~600ms** under high traffic.
- Designed and implemented a distributed Redis + Lua rate limiter (atomic counters, TTL expiration) enforcing per-client throttling and eliminating API overload and Redis memory growth.
- Architected and provisioned cloud infrastructure end-to-end (Terraform, Kubernetes, Docker, CI/CD) spanning networking, compute, IAM, load balancing, autoscaling, and monitoring; reduced AWS cost by **~50%** with no performance trade-off.
- Cut Docker build times by **60–90%** and manual deployment effort by **70%**, enabling zero-downtime production releases.
- Migrated Redux to Redux Toolkit, cutting boilerplate and lifting frontend state-management performance by **~30%**.

Software Developer | Volumtree Technology Solutions Pvt. Ltd — Chandigarh, India *May 2022 – Oct 2022*

- Re-architected a Salesforce data pipeline processing **~1.5M records**, cutting execution time from days to **under 5 minutes**.
- Optimized a React application (memoization, debouncing/throttling, lazy loading, code splitting, tree shaking, virtualization), cutting page load time by **30%** and lifting UI responsiveness by **50%**.
- Architected a hybrid SQL + NoSQL data layer for real-time chat, boosting throughput while keeping latency low under high concurrency.
- Migrated a monolithic application to microservices, improving scalability and fault isolation.
- Re-engineered product rating aggregation to prevent server overload, cutting server load by **70%** and improving customer-facing latency.

Software Engineer | RedoQ Ltd (India Office) — Kolkata, India *Oct 2020 – Apr 2022*

- Built scalable restaurant management systems adopted by **5,000+ businesses**, cutting customer queue times by **50%**.
- Designed systems handling **100K+ RPS** through caching strategies and optimized database access patterns.
- Migrated legacy Laravel services to Node.js, improving backend performance and long-term maintainability.

- Implemented gRPC-based inter-service communication for internal microservices, improving service-to-service call performance over REST.
- Re-architected a monolith from vertical to horizontal scaling using AWS EC2 Auto Scaling and stateless Node.js services, increasing scalability and reducing downtime.
- Designed a pre-aggregation pipeline for daily/monthly/yearly analytics, eliminating heavy runtime queries and cutting database load.
- Refactored the rating/review system from bulk updates to single-write patterns, significantly reducing database load.

Software Engineer | II Intlum Technology — Kolkata, India *Oct 2019 – Sep 2020*

- Designed and built RESTful and GraphQL APIs supporting core product operations, with Swagger documentation for front-end teams.
- Built reusable React components (buttons, modals, forms), reducing development time on new features.
- Replaced offset-based pagination with cursor-based pagination, improving query performance at scale.

EDUCATION

Bachelor of Technology — Electronics and Communication Engineering

MAKAUT University, West Bengal, India | Jun 2014 – Jul 2018